

CLASS 11TH MATHEMATICS

ANNUAL BOARD PAPER

Series: 11thARNKD(W/Z) JKLUT-25-1207-A Time: 3.0 Hours

Max. Marks: 80 Marks

Note: Attempt any 68 Marks out of 80 Marks. | All sections are compulsory as per given internal choices.

SECTION A: OBJECTIVE / MULTIPLE CHOICE QUESTIONS

1 Mark each

- Q1. The roster form of the set $A = \{x : x \text{ is an integer and } -3 \leq x < 7\}$ is:
(A) $[-3, 7]$ (B) $(-3, 7]$
(C) $\{-3, -2, -1, 0, 1, 2, 3, 4, 5, 6\}$ (D) $[-3, 7]$
- Q2. If $n(A) = 2$ and $n(B) = 3$, then the total number of distinct relations from set A to set B is:
(A) 64 (B) 128 (C) 256 (D) None of these
- Q3. If $\sin(\theta) = 1/2$ and θ lies in the 2nd Quadrant, then $\cos(\theta)$ is equal to:
(A) $\sqrt{3}/2$ (B) $-\sqrt{3}/2$ (C) $-1/2$ (D) None of these
- Q4. If $z = 2 + i$, then the modulus $|z|$ is equal to _____.
- Q5. Combinatorial identity: ${}^nC_r = {}^nC_{(n-r)}$. (State True or False)
- Q6. The coordinates of the centre of the circle $x^2 + y^2 - 6x + 4y - 5 = 0$ are:
(A) (3, -2) (B) (-3, 2) (C) (-3, -2) (D) (3, 2)
- Q7. The length of the transverse axis of the hyperbola $x^2/16 - y^2/9 = 1$ is:
(A) 8 (B) 6 (C) 10 (D) None of these
- Q8. The algebraic solution set of the linear inequality $24x > 100$ for real x is:
(A) $(25/6, \infty)$ (B) $(-25/6, \infty)$ (C) $(-\infty, 25/6)$ (D) None of these
- Q9. Define primary data in statistical investigations.
- Q10. If A is an impossible event in sample space S, then $P(A) = 0$. (State True or False)

SECTION B: VERY SHORT ANSWER TYPE QUESTIONS

2 Marks each

- Q11. If the ordered pairs are equal: $(x/3 + 1, y - 2/3) = (5/3, 1/3)$, find the values of x and y.
- Q12. Find the range of the real-valued function $f(x) = x^2 + 2$, where x is a real number.
- Q13. Without using Pythagoras theorem, show that the points (4, 4), (3, 5), and (-1, -1) form the vertices of a right-angled triangle using coordinate slopes.
- Q14. Reduce the linear equation $3x + 2y - 12 = 0$ into intercept form $(x/a + y/b = 1)$ and determine its intercepts on the coordinate axes.
- Q15. Find the standard equation of the parabola having its focus at (6, 0) and directrix line $x = -6$.
- Q16. Find the exact trigonometric value of $\sin(765 \text{ degrees})$.

- Q17.** A playing card is randomly selected from a well-shuffled standard pack of 52 cards. Find the probability that the card drawn is an ace.
- Q18.** Find the mean and variance of the numerical dataset: 6, 7, 10, 12, 13, 4, 8, 12.
- Q19.** Expand completely using Binomial Theorem: $(x - 1/x)^5$.
- Q20.** For what real values of x are the numbers $-2/7$, x , and $-7/2$ in Geometric Progression (G.P.)?

SECTION C: SHORT ANSWER TYPE QUESTIONS

4 Marks each

- Q21.** Let $A = \{1, 2, 3, 5\}$ and $B = \{4, 6, 9\}$. Define a relation R from A to B by $R = \{(x, y) : \text{the difference between } x \text{ and } y \text{ is odd; } x \text{ in } A, y \text{ in } B\}$. Write R in roster form and state its domain and range.
- Q22.** If $\sin(x) = 3/5$, where x lies in the second quadrant, find the values of the other five trigonometric functions.
- Q23.** For any two complex numbers z_1 and z_2 , prove algebraically that:
 $\text{Re}(z_1 * z_2) = \text{Re}(z_1) * \text{Re}(z_2) - \text{Im}(z_1) * \text{Im}(z_2)$.
- Q24.** Solve the system of linear inequalities and represent the solution set graphically on a real number line:
 $3x - 7 > 2(x - 6)$ and $6 - x > 11 - 2x$.
- Q25.** Find the standard equation of the ellipse whose major axis lies on the x -axis and passes through the points $(4, 3)$ and $(6, 2)$.
- Q26.** Evaluate the algebraic limit: $\lim_{x \rightarrow 3} (x^4 - 81) / (2x^2 - 5x - 3)$.
- Q27.** Find the derivative of the function $f(x) = x^3 - 27$ with respect to x from first principles (ab-initio method).
- Q28.** Find the positive integer value of n such that: ${}^n P_5 = 42 * {}^n P_3$, where $n > 4$.

SECTION D: LONG ANSWER TYPE QUESTIONS

6 Marks each

- Q29.** Prove the trigonometric identity:
 $(\cos 4x + \cos 3x + \cos 2x) / (\sin 4x + \sin 3x + \sin 2x) = \cot 3x$.
 --- OR ---
 Prove the trigonometric identity:
 $\cos(3\pi/2 + x) * \cos(2\pi + x) * [\cot(3\pi/2 - x) + \cot(2\pi + x)] = 1$.
- Q30.** Find the sum to n terms of the sequence: $8 + 88 + 888 + 8888 + \dots$ to n terms.
 --- OR ---
 If p -th, q -th, and r -th terms of a Geometric Progression (G.P.) are a , b , and c respectively, prove that:
 $a^{(q-r)} * b^{(r-p)} * c^{(p-q)} = 1$.
- Q31.** If E and F are events such that $P(E) = 1/4$, $P(F) = 1/2$, and $P(E \text{ and } F) = 1/8$, calculate:
 (i) $P(E \text{ or } F)$
 (ii) $P(\text{not } E \text{ and not } F)$
 --- OR ---
 Calculate the Mean Deviation about Median for the following grouped frequency distribution data:
 Marks: 0-10 10-20 20-30 30-40 40-50 50-60
 No. of Girls: 6 8 14 16 4 2